

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, SPRING SEMESTER 2022-2023

MACHINE LEARNING

Time allowed: 2 hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer All Questions

Only silent, self contained calculators with a Single-Line Display or Dual-Line Display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

In the K-Nearest Neighbour (kNN) model, the class of a test point X is determined by the most common class among its K nearest neighbours in the training set. Given a test point X , the algorithm finds the K nearest training samples using a distance measure, such as the Euclidean distance, and assigns X to the class obtained by majority voting. kNN is an instance-based learning method. The value of K is typically selected using N -fold cross validation to minimise the cross validation error: a small K may lead to overfitting, while a large K may cause underfitting.

- (i) A learning algorithm first constructs a decision tree of fixed depth k using standard splitting criteria. For each leaf, a Naïve Bayes classifier is trained on the subset of training samples reaching that leaf using the remaining $n-k$ variables. During prediction, an input is routed to a leaf and classified by the Naïve Bayes model stored there.
- (ii) No. Each leaf corresponds to a different subset of training data defined by the decision tree path, leading to different class priors and conditional probability estimates, and hence different Naïve Bayes classifiers.
- (iii) The value of k can be selected using cross validation by treating it as a hyperparameter and choosing the depth that minimises validation error.

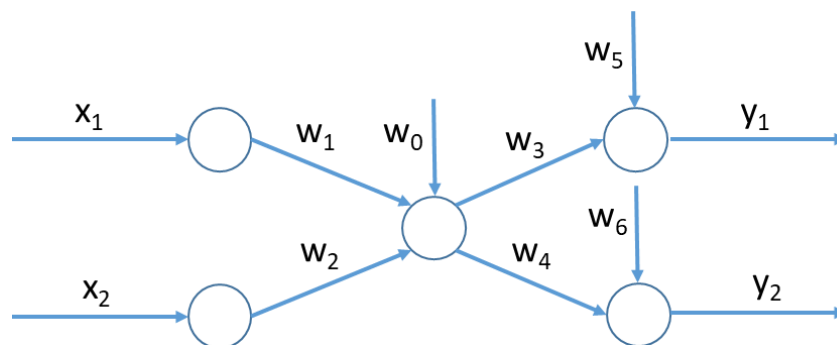
Question 1:

- (a) What is the purpose of kNN ? What is the effect of different k values (small or large)? How to use cross validation to tune the parameters of a learning algorithm, e.g. k in kNN , to a given application? [10 marks]
- (b) Suppose you are to argument a decision tree by adding a Naïve Bayes classifier at each leaf. That is, consider learning a decision tree of depth k , where k is smaller than the number of variables n ; each leaf contains not a class, but a Naïve Bayes classifier, which is based on the $n-k$ variables not appearing on the path to that leaf. (i) Briefly outline a learning algorithm for this representation, assuming that k is a fixed parameter. (ii) Will this result in having the same Naïve Bayes Classifier at each leaf? Why or why not? (iii) Briefly describe a plausible means of selecting a value for k ? [12 marks]

文本

Question 2:**[20 marks]**

- (a) Considering the following feed-forward neural network.



The operation of the network is described by the following equations:

$$y_1 = (w_0 + w_1x_1 + w_2x_2)w_3 + w_5$$

$$y_2 = (w_0 + w_1x_1 + w_2x_2)w_4 + w_6$$

Suppose the network tries to generate the output whose sum equals to the sum of the input, the cost function of the network, E , can be defined as follows:

$$E = \frac{(x_1 + x_2 - y_1 - y_2)^2}{2}$$

To minimize the error, we can use gradient descent technique. Derive the gradient descent equations for updating w_0 , w_1 , w_2 , w_3 , w_4 , w_5 , and w_6 , respectively. An answer without the working process will not receive any marks.

[16 marks]

- (b) In back-propagation algorithm, we need to decide the learning rate. Briefly explain how the learning rate affects the convergence of the algorithm?

The learning rate controls the step size of weight updates in the back-propagation algorithm. If the learning rate is too small, the convergence is very slow because the weights are updated by very small amounts. If the learning rate is too large, the algorithm may overshoot the minimum, causing oscillation or even divergence. An appropriate learning rate allows the algorithm to converge efficiently and stably to a minimum of the error function. [4 marks]

If the learning rate is sufficiently small, the algorithm can converge stably to a (local) minimum of the error function.

Question 3:

- (a) You will simulate the k-means algorithm to identify two clusters in the following $\{x,y\}$ pairs.

Data#	x	y
1	1.90	0.97
2	1.76	0.84
3	2.32	1.63
4	2.31	2.09
5	1.14	2.11
6	5.02	3.02
7	5.74	3.84
8	2.25	3.47
9	4.71	3.60
10	3.17	4.96

Suppose you are given the initial assignment cluster center as {cluster-1: Data#1}, {cluster2: Data#10}, that is, the first data point is used as the first cluster center and the last data point is used as the second cluster center. Assume the k-means algorithm uses Euclidean distance. Please simulate the k-means algorithm for iterations. What are the cluster assignments after ONE iteration? What are the cluster assignments until convergence? (Fill in the table below)

[10 marks]

Data#	Cluster Assignment after ONE iteration	Cluster Assignment after convergence
1	1	1
2	1	1
3	1	1
4	1	1
5	1	1
6	2	2
7	2	2
8	2	2
9	2	2
10	2	2

- (b) Let Y and Z be random variables, where $0 < a < b < 1$, and the below table lists:

Y	Z	P(Y)	P(Z)	P(Y,Z)
0	0	1-a	b	b-a
0	1	1-a	1-b	1-b
1	0	a	b	a
1	1	a	1-b	0

Is Y and Z independent? Why or why not? Use calculation to justify your answer.

[6 marks]

- (c) Suppose you are trying to learn regression parameters for a dataset which you know was generated from a polynomial of a certain degree, but you do not know what this degree

Since the true data-generating function is a 4th-degree polynomial with Gaussian noise, a 3rd-degree model cannot represent the x^4 term and will therefore suffer from systematic underfitting (high bias), leading to poorer generalisation. A 5th-degree model contains the true 4th-degree model as a special case (it can learn the extra coefficient close to zero), giving lower bias. Although the 5th-degree model may have higher variance, with 100 training samples and only one extra degree, it is more likely to achieve a better bias-variance trade-off and thus fit the test data better.

COMP3055

is. Assume that the data was actually generated from a polynomial of degree 4 with some added Gaussian noise, that is $y = w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + \epsilon$, where $\epsilon \sim N(0,1)$. For training you have 100 $\{x,y\}$ pairs and for testing you are using an additional set of 100 $\{x,y\}$ pairs. Since you do not know the degree of the polynomial, you learn two models from the data. Model-i learns parameters for a polynomial of degree 3 and Model-j learns parameters for a polynomial of degree 5. Which of these two models is likely to fit the test data better? Justify your answer.

[6 marks]

(d) Suppose you are constructing a decision tree for a dataset with continuous attributes. For such attributes, you can use threshold splits to determine the best partition for a set of vectors. Assume you are at the root and you have n vectors, all with different (continuous) values for attribute x_1 .

(i) Assume you would like to use binary splits. For such splits, you need to choose a value a and split the data by propagating all vectors with $x_1 < a$ to the left and those with $x_1 > a$ to the right. For any value of a that you consider, at least one vector would be assigned to each of the two branches of the split. How many values of a do you need to consider?

[3 marks]

(ii) Assume you would like to use three way splits. For such splits, you need to choose value a and b such that $a < b$ and split the data into three sets: $x_1 < a$, $a \leq x_1 < b$, $x_1 \geq b$. Again for any value of a and b that you consider, at least one vector would be assigned to each of the three branches of the split. How many $\{a,b\}$ do you need to consider?

[3 marks]

Question 4:

[30 marks]

(a) Briefly describe linear separability and non-linear separable problem in machine learning.

A classification problem is linearly separable if there exists a linear decision boundary, such as a line or hyperplane, that can perfectly separate data points of different classes. In contrast, a non-linearly separable problem is one where no linear decision boundary can separate the classes correctly, and a nonlinear boundary or feature transformation is required.

[4 marks]

(b) Suppose we use SVM classifier with 5-fold Cross Validation for a classification problem and the confusion matrix is shown in the following table. State the total number of examples, correctly classified examples, and what the recall, precision and F1 for class C are. An answer without the working process will not receive any marks.

Both Perceptron and ADALINE are linear classifiers that use a decision plane to separate two classes. However, they differ in their training techniques. The Perceptron uses a hard threshold activation function and updates its weights only when a misclassification occurs, aiming to find any separating hyperplane for linearly separable data. As a result, when the data are not linearly separable, the Perceptron algorithm fails to converge. In contrast, ADALINE uses a linear activation function and applies the delta rule to minimise the mean squared error using gradient descent. By optimising a continuous error function, ADALINE provides more stable training and can find a best-fit solution even when the data are not perfectly linearly separable.

[8 marks]

	Predicted		
Actual	A	B	C
A	50	10	20
B	10	60	20
C	30	20	50

(c) Both Perceptron and ADLINE use a decision plane to separate two classes. Compare and contrast these training techniques.

- (d) LSTM is an excellent model for solving sequence-to-sequence problems. It contains multiple components such as forget gate, input gate, output gate, etc. Briefly describe the functionality of forget gate and input gate.

[6 marks]

- (e) Actor-Critic method is one of the reinforcement learning algorithms. Briefly describe the main idea of Actor-Critic. You may draw a diagram to help illustrate your understanding.

[6 marks]

d)

1. Forget Gate

-The first gate determines whether to carry over the history or forget it

-Called "forget" gate.

-Actually, determine how much history to carry over.

-The memory C and hidden state h are distinguished.

2. Input Gate

-The second gate has two parts

-A tanh unit determines if there is something new or interesting in the input.

-A gate decides if it is worth remembering.

Forget Gate:

The forget gate determines how much information from the previous cell state should be carried over or forgotten. Rather than making a binary decision, it controls the proportion of historical information to retain, allowing the LSTM to discard irrelevant or outdated memory. In LSTM, the cell state C and hidden state h are treated separately.

Input Gate:

The input gate consists of two components. A tanh unit generates candidate new information from the current input, while a gating mechanism decides how much of this new information is worth remembering and should be written into the cell state.

- e) The Actor-Critic method combines policy-based and value-based reinforcement learning. It consists of two components: the actor and the critic. The actor is responsible for selecting actions according to a policy, while the critic evaluates the action by estimating a value function and computing a temporal-difference (TD) error. The critic's feedback is then used to update the actor's policy, enabling the agent to learn better actions over time.

